

Scalability & Future Plan

Date	4 July 2026
Team ID	
Project Name	A Comprehensive Measure Of Well-Being
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited Input Variables	Only a few indicators are used for prediction	Medium
2	Local Deployment	Accessible only locally via Ngrok.	High
3	Static Dataset	Model doesn't update with real-time global data.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Handles single-user requests locally.	Deploy on Cloud (AWS/Render) for multi-user support.
2	Data Storage	Uses CSV files for historical data.	Integrate a SQL Database (PostgreSQL) for scalability.
3	Performance	Basic model inference.	Optimize model using caching and batch processing.
4	Security	Minimal input validation.	Add OAuth2 authentication and API rate limiting.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Real-time Data API Integration	Next 1 Month	Up-to-date HDI predictions.

Phase 2	Interactive Data Visualization Dashboard	Next 2 Months	Better insights for users.
Phase 3	Mobile App Development	Next 4 Months	Wider accessibility.
Phase 4	Multilingual Support & Global Scaling	Next 6 Months	Global user engagement.